

DS 642: Applications of Parallel Computing

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Week 14:

Final Project and Concluding Thoughts

Project Report

Project Goal: understand performance

- Provide description of your problem.
 - A description of the design choices that you tried and how did they affect the performance.
- Describe performance analysis, including:
 - Serial tuning
 - Strong and weak scaling study
 - Speedup plots
 - Profiling of communication and computation
 - Parallelization strategy (communication, synchronization, etc.)
 - Comparison to analytical models
- Future work/direction
- The organization of the code:
 - Provide comments in the code.
 - Describe error handling and provide tests.

Goals

Understand reasons about code performance and optimization:

- Hardware
 - Tune existing codes for modern hardware
- Software
 - Apply good software practices
- Algorithm
 - Good models that are also easy to program

Many factors to consider

- Pipelining, concurrent execution, and vectorization
- Memory hierarchy and the cost of data movement
- Communication costs (latency and bandwidth)
- Synchronization overheads